Detailed Analysis Report

1. Dataset Overview

Total Observations: **911**

Total Variables: **39**

2. Level-0 Operation

Pass Count: **834**
Hold Count: **77**
Pass Percentage: **91.5%**

Inference

Approximately ninety-one point five percent (91.5%) of the collected observations successfully surpass the initial threshold, or Level-0. In this instance, a total of 834 observations exceeded Level-0, whereas 77 observations were retained. It should be noted that Level-0 seldom hinders the generation of recommendations and primarily serves as an introductory filtering stage.

Conclusion

A grand total of 834 observations have successfully surpassed Level-0, whereas 77 observations remain in hold status. These findings encapsulate the outcomes of the Level-0 screening process.

3. Level-1 Behaviour

Hold Count: **525**
Go To Level 2: **307**
Hold Percentage: **62.9%**
Level 2 Percentage: **36.8%**

Inference

Among the observations clearing Level-0 scrutiny, 62.9% are retained at Level-1. In aggregate, 525 observations are retained at Level-1. A sum of 307 observations advance to Level-2, constituting 36.8% of Level-0 clearances. Level-1 serves as an intermediary decision stratum.

Conclusion

The analysis conducted at Level-1 encompasses a total of 525 observations, out of which 307 are forwarded to Level-2. This report presents an overview of the decisions made at Level-1.

4. Level-2 Behaviour

Raise Count: **234**
Hold Count: **73**

Evaluation Percentage: **33.7%**

Inference

A grand total of 234 instances experience an adjustment action designated as Level-2. Within this number, there are 73 instances that remain stationary at Level-2. It is observed that a mere 33.7% of the total observations necessitate an evaluation at Level-2.

Conclusion

A grand total of 234 observations have been subjected to raise setpoint actions, while 73 observations have undergone hold actions at Level-2. This evaluation at Level-2 encompasses approximately 33.7% of the entire observation pool. These findings serve as a synopsis of the Level-2 decision's outcome.

5. Chiller 1 Analysis

Hold Percentage: **60.4%**
Raise Percentage: **39.4%**
Reduce Percentage: **0.2%**

Inference

For Chiller 1, approximately sixty percent of the suggested actions pertain to maintenance as 'hold.' Approximately forty percent of the recommended actions concern elevating setpoint settings. Less than one percent involve decreasing setpoint settings.

Conclusion

The primary suggestion put forth concerning Chiller 1 is maintenance at a standstill. In summary, a grand total of 536 suggestions have been produced in relation to Chiller 1.

6. Chiller 2 Analysis

Hold Percentage: **48.7%**
Raise Percentage: **51.1%**
Reduce Percentage: **0.2%**

Inference

In reference to Chiller 2, it is noted that 48.7% of the suggestions pertain to maintenance as hold actions. Conversely, approximately 51.1% of the recommendations entail an increase in the setpoint. A minimal percentage of 0.2% of the suggestions are intended for a reduction in the setpoint.

Conclusion

The primary suggestion issued for Chiller 2 pertains to an adjustment in setpoint, specifically referred to as 'raise_setpoint.' In the course of generation, a grand total of 450 recommendations were produced for Chiller 2.

7. Chiller Comparison

CH1 Hold: **324**
CH2 Hold: **219**
CH1 Raise: **211**
CH2 Raise: **230**

Inference

Chiller Number One has been issued a total of 324 hold recommendations and 211 raise recommendations. In the case of Chiller Number Two, it has received 219 hold recommendations and 230 raise recommendations.

Conclusion

The primary suggestion for Chiller 1 is maintenance at its current setting. For Chiller 2, the principal advice is to increase the setpoint. This comparison underscores the disparity in recommended action between the two chillers.

8. Chiller 1 Current Operating Setpoint

Mean: **9.0°C**
Median: **9.0°C**
Min: **4.5°C**
Max: **9.0°C**

The existing operational parameters for Chiller 1 span between 4.5°C and 9.0°C. The average (mean) and median operating temperatures, respectively, for Chiller 1 are both recorded as 9.0°C. This data provides an overview of the observed operational behavior of Chiller 1.

9. Chiller 1 Recommended Setpoint

Mean: **9.24°C**
Median: **9.3°C**
Min: **9.0°C**
Max: **9.5°C**

Recommended chiller 1 setpoints span between 9.0°C and 9.5°C, with a mean value of 9.24°C and a median value of 9.3°C. These figures constitute the suggestions derived from the optimization framework.

10. Chiller 2 Current Operating Setpoint

Mean: **9.0°C**
Median: **9.0°C**
Min: **9.0°C**
Max: **9.0°C**

The current operating setpoints for Chiller 2 span a range from 9.0°C to 9.0°C. The average (mean) and median operating setpoint for this chiller are both 9.0°C. These statistics provide an overview of the operational behavior exhibited by Chiller 2.

11. Chiller 2 Recommended Setpoint

Mean: **9.29°C**
Median: **9.3°C**
Min: **9.0°C**
Max: **9.5°C**

The suggested chiller 2 temperature settings span between 9.0°C and 9.5°C. The average (mean) of these suggested setpoints amounts to 9.29°C, while the middle value (median) is 9.3°C. These values signify the temperature recommendations produced by the optimization framework.

12. Key Difference Between the Two Chillers

The primary suggestion for Chiller 1 is to maintain its current state. For Chiller 2, the principal advice is to increase the setpoint. These recommendation patterns suggest distinct operational variations between the two chillers.

13. Setpoint Reduction Opportunities are Extremely Rare

Chiller 1 Reduce Count: **1**
Chiller 2 Reduce Count: **1**

14. Hierarchical Decision Structure

911 observations
↓
Level-0
↓
834 pass
↓
Level-1
↓
525 hold
307 → Level-2
↓
234 raise setpoint
73 hold

Overall Interpretation

The data set encompasses a total of 911 instances. Ninety-one point five percent (91.5%) of the observations are associated with Level-0 passes. In regards to Chiller 1, the predominant advice is to maintain its current setting. For Chiller 2, the most frequent recommendation is to adjust the setpoint upward. These findings provide an overview of the general performance of the recommendation system.

Overall Plant Behaviour

The primary suggestion for Chiller 1 is to maintain its current state. For Chiller 2, the principal advice is to increase the setpoint. These recommendation patterns suggest distinct operational variations between the two chillers.

Main Takeaway

Level-Zero attains a success rate of 91.5%. For Chiller Number One, the principal advisement is to maintain its current state. With respect to Chiller Number Two, the leading guidance is to elevate its setpoint. These findings encapsulate the most prominent results derived from the recommendation phase.